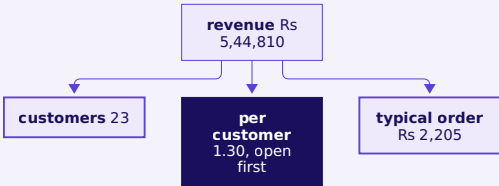


The revenue tree

Kalpa Retail, Week 1 Monday. Revenue is a tree of metrics that multiply, every number carries its definition, and each chapter's trap below is a plausible wrong number with the check that catches it. The retail story's formulas are on the domain card.

1 The revenue tree, with Monday's numbers on it



Crux. Revenue is customers times orders per customer times average order value; the identity $23 \times (30 / 23) \times (\text{Rs } 5,44,810 / 30)$ lands on Rs 5,44,810.

2 Ch 1, which total is "sales"

Wrong number	Rs 5,44,810 called sales, with 4 cancelled store orders inside it.
Check	Count by status first: 21 delivered, 5 returned, 4 cancelled.
Fix	Not cancelled is Rs 5,35,760 on 26 orders; delivered is Rs 5,20,790 on 21.

Crux. Name the definition before the number: booked, "not cancelled", or delivered.

3 Ch 2, a fraction from two reports

Wrong number	AOV Rs 25,943: booked rupees over delivered orders.
Check	AOV x the 30 orders the revenue covers gives Rs 7,78,300, not Rs 5,44,810.
Fix	Rs 18,160 booked, or Rs 24,800 delivered, each named.

Crux. Build every fraction on one definition, and check that it multiplies back.

4 Ch 3, rows counted as customers

Wrong number	30 customers, so $30 / 30 = 1.00$ and "nobody comes back".
Check	<code>len(rows)</code> against <code>len(set(ids))</code> : 30 against 23.
Fix	23 customers, 1.30 orders each, 7 came back and 16 bought once.

Crux. Count customers by their id, never by the rows.

5 Ch 4, the mean sold as typical

Wrong number	A first order valued at the mean, Rs 18,160.
Check	Orders above the mean: 1 of 30. Sort and read the top.
Fix	Median Rs 2,205 booked, Rs 2,060 delivered; the top order gets its own line.

Crux. Report the median when one order can move the mean, and say why.

6 Ch 5, lifts added instead of multiplied

Wrong number	Customers +10 percent and frequency +10 percent called 20 percent.
Check	Recompute through the tree: $1.10 \times 1.10 = 1.21$.
Fix	Rs 6,59,220, not Rs 6,53,772. On the 29 everyday orders the plan needs 3.3 customers or 4.35 orders: open frequency.

Crux. Lifts multiply along the tree: two 10 percent lifts make 21 percent.

7 Ch 6, one-time buyers called lost

Wrong number	16 of 23 bought once, so "70 percent are lost".
Check	Median repeat gap 45 days; 9 of the 16 bought in the last 45.
Fix	7 came back, 7 had time and did not, 9 are too recent to judge.

Crux. A one-time buyer is not a lost customer until they have had time to come back.

8 What goes to Meera

"23 customers, 1.30 orders each, a typical order of Rs 2,205; 7 came back and 9 are too recent to judge, so open frequency first and hold the Rs 12 crore for Tuesday's two quarters."

Crux. One window shows the shape of revenue; only two windows show which branch moved.

THE WORDS ON THIS SHEET

- Revenue tree** Revenue drawn as the metrics that multiply into it, less discounts
- Booked revenue** Every order placed in the window, whatever happened to it later
- Denominator** What a metric is divided by, said before it is computed
- Delivered revenue** Only the orders that reached the customer and stayed

- Identity check** Multiplying a tree's branches back to see they land on the total
- Distinct customers** Each customer id counted once, however many orders it placed